

BVT2

Physical Chemistry

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

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How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	_____	

Items

1.

A process engineer at a plant with 200 employees studies a reaction with $\Delta H = -50 \text{ kJ/mol}$ and $\Delta S = -100 \text{ J/(mol}\cdot\text{K)}$ at 298 K. The reactor was installed in 2011. Is the reaction spontaneous at this temperature?

- A. Yes; $\Delta G \approx -79.8 \text{ kJ/mol}$.
- B. No, because a reaction with negative ΔS is never spontaneous.
- C. No; $\Delta G \approx +29,750 \text{ kJ/mol}$.
- D. Yes; $\Delta G \approx -20.2 \text{ kJ/mol}$.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

2. [TRANSFER — reversed stem / different variable]

For a reaction with $\Delta H = -50 \text{ kJ/mol}$ and $\Delta S = -100 \text{ J/(mol}\cdot\text{K)}$, over what temperature range is it spontaneous?

- A. Above 500 K
- B. Below 0.5 K
- C. At all temperatures, since ΔH is negative.
- D. Below 500 K

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

3.

At a foundry that runs three furnaces, a chemist has these data: $\text{C} + \text{O}_2 \rightarrow \text{CO}_2$, $\Delta H = -394 \text{ kJ}$; $\text{CO} + \frac{1}{2}\text{O}_2 \rightarrow \text{CO}_2$, $\Delta H = -283 \text{ kJ}$. What is ΔH for $\text{C} + \frac{1}{2}\text{O}_2 \rightarrow \text{CO}$?

- A. -394 kJ
- B. $+111 \text{ kJ}$
- C. -677 kJ
- D. -111 kJ

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

4.

In a pharmaceutical lab whose new spectrometer cost \$80,000, doubling [A] while holding [B] constant makes the initial rate four times as fast. What is the reaction order in A?

- A. Fourth order
- B. Second order
- C. Half order
- D. First order

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

5. [TRANSFER — reversed stem / different variable]

Rate = $k[A]^2[B]$. If [A] is tripled and [B] is cut in half, by what factor does the rate change?

- A. 9
- B. 4.5
- C. 18
- D. 1.5

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

6.

A catalyst supplier with offices in three countries sells a catalyst for a reversible reaction. Once the catalyst is added, which statement is true?

- A. K increases, so more product forms.
- B. Activation energy drops and equilibrium is reached sooner; ΔH and K do not change.
- C. It makes any nonspontaneous reaction spontaneous.
- D. ΔH becomes more negative.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

7.

At an ammonia plant that supplies fertilizer to 12 states, the reaction $N_2 + 3H_2 \rightleftharpoons 2NH_3$ has $\Delta H = -92$ kJ. Operators raise the temperature. What happens at the new equilibrium?

- A. It shifts toward N_2 and H_2 , and K decreases.
- B. It shifts toward NH_3 , because higher temperature speeds the reaction.
- C. No shift, since temperature affects only the rate.
- D. It shifts toward NH_3 , because heat is a reactant.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

8.

For $A \rightleftharpoons B$, $K = 0.50$ at 450°C . A technician who just finished safety training measures $[A] = 2.0\text{ M}$ and $[B] = 0.40\text{ M}$. Which way will the reaction proceed?

- A. No net change; the system is at equilibrium.
- B. In reverse, because $Q = 5.0$ is larger than K .
- C. Forward, toward B, because $Q < K$.
- D. In reverse, toward A, because $Q < K$.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

9. [TRANSFER — reversed stem / different variable]

A reaction mixture has $Q = 12$ and is observed to shift toward reactants. What must be true of K ?

- A. $K = 12$
- B. $K > 12$
- C. $K < 12$
- D. K must be negative.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

10.

A battery startup that raised \$5 million builds a zinc-copper galvanic cell. Standard reduction potentials are $\text{Zn}^{2+}/\text{Zn} = -0.76\text{ V}$ and $\text{Cu}^{2+}/\text{Cu} = +0.34\text{ V}$. What is E°_{cell} ?

- A. $+0.42\text{ V}$
- B. -0.42 V
- C. $+1.10\text{ V}$
- D. -1.10 V

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

11. [TRANSFER — reversed stem / different variable]

A cell has $E^\circ_{\text{cell}} = +0.50\text{ V}$ and transfers 2 electrons per reaction ($F = 96,485\text{ C/mol}$). What is ΔG° ?

- A. About -96.5 kJ
- B. Positive, since any cell under 1 V is nonspontaneous.
- C. About $+96.5\text{ kJ}$
- D. About -48.2 kJ

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

12.

A tire filled at a service station that opened in 1985 reads 200 kPa at 27 °C. After a highway drive the air inside warms to 57 °C with no change in volume. What is the new pressure?

- A.** 220 kPa
- B.** About 422 kPa
- C.** About 182 kPa
- D.** 230 kPa

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer D Concept 1

A process engineer ~~at a plant with 200 employees~~ studies a reaction with $\Delta H = -50 \text{ kJ/mol}$ and $\Delta S = -100 \text{ J/(mol}\cdot\text{K)}$ at 298 K. ~~The reactor was installed in 2011.~~ Is the reaction spontaneous at this temperature?

Target word: sign of ΔG at 298 K **Decides:** $\Delta G = -50 - 298(-0.100) = -20.2 \text{ kJ/mol}$.

A	Inverted operation	Subtracts $T\Delta S$ with the wrong sign.
B	Absolute language	A negative ΔH can outweigh a negative ΔS at low temperature.
C	Dropped element	Drops the J-to-kJ conversion for ΔS .
D	Correct	$-50 + 29.8 = -20.2$, which is negative.

2. Answer D Concept 1 • transfer

For a reaction with $\Delta H = -50 \text{ kJ/mol}$ and $\Delta S = -100 \text{ J/(mol}\cdot\text{K)}$, over what temperature range is it spontaneous?

Target word: solve for the temperature boundary **Decides:** $\Delta G = 0$ at $T = \Delta H/\Delta S = 500 \text{ K}$.

A	Inverted operation	Above 500 K the $-T\Delta S$ term makes ΔG positive.
B	Confusion set	Mixes kJ and J units in $\Delta H/\Delta S$.
C	Dropped element	Drops the entropy term.
D	Correct	Below 500 K the $-\Delta H$ term dominates, so $\Delta G < 0$.

3. Answer D Concept 2

~~At a foundry that runs three furnaces,~~ a chemist has these data: $\text{C} + \text{O}_2 \rightarrow \text{CO}_2$, $\Delta H = -394 \text{ kJ}$; $\text{CO} + \frac{1}{2}\text{O}_2 \rightarrow \text{CO}_2$, $\Delta H = -283 \text{ kJ}$. What is ΔH for $\text{C} + \frac{1}{2}\text{O}_2 \rightarrow \text{CO}$?

Target word: ΔH of target reaction **Decides:** Keep step 1, reverse step 2.

A	Dropped element	Uses only the first step.
B	Confusion set	Reverses the wrong step.
C	Inverted operation	Adds both steps without reversing the second.
D	Correct	$-394 + 283 = -111$.

4. Answer B Concept 3

In a pharmaceutical lab ~~whose new spectrometer cost \$80,000~~, doubling [A] while holding [B] constant makes the initial rate four times as fast. What is the reaction order in A?

Target word: order in A **Decides:** $2^m = 4$.

A	Wrong stem number	Uses the rate factor as the order.
B	Correct	$2^2 = 4$.
C	Inverted operation	Solves $4^m = 2$ instead of $2^m = 4$.
D	Dropped element	Assumes rate is simply proportional to [A].

5. Answer B Concept 3 · transfer

Rate = $k[A]^2[B]$. If [A] is tripled and [B] is cut in half, by what factor does the rate change?

Target word: rate factor **Decides:** $3^2 \times 0.5$.

A	Dropped element	Drops the change in B.
B	Correct	$9 \times 0.5 = 4.5$.
C	Inverted operation	Doubles for B instead of halving.
D	Confusion set	Ignores the exponent on A: 3×0.5 .

6. Answer B Concept 4

A catalyst supplier ~~with offices in three countries~~ sells a catalyst for a reversible reaction. Once the catalyst is added, which statement is true?

Target word: what a catalyst changes **Decides:** Lower activation energy; equilibrium position unchanged.

A	Adjacent category	Confuses speed with equilibrium position.
B	Correct	A catalyst changes the path, not the thermodynamics.
C	Absolute language	A catalyst cannot change ΔG .
D	Confusion set	ΔH is a state function; the catalyst does not change it.

7. Answer A Concept 5

At an ammonia plant ~~that supplies fertilizer to 12 states~~, the reaction $N_2 + 3H_2 \rightleftharpoons 2NH_3$ has $\Delta H = -92$ kJ. Operators raise the temperature. What happens at the new equilibrium?

Target word: effect of heat on an exothermic equilibrium **Decides:** Heat acts as a product; raising T shifts left.

A	Correct	The system absorbs added heat by running the endothermic reverse direction.
B	Confusion set	Confuses faster rate with equilibrium position.
C	Adjacent category	Temperature is the one change that alters K.
D	Inverted operation	In an exothermic reaction heat is a product.

8. Answer C Concept 6

For $A \rightleftharpoons B$, $K = 0.50$ at 450°C . A technician ~~who just finished safety training~~ measures $[A] = 2.0$ M and $[B] = 0.40$ M. Which way will the reaction proceed?

Target word: direction of shift **Decides:** $Q = 0.40 / 2.0 = 0.20 < 0.50$.

A	Dropped element	Skips the Q calculation.
B	Inverted operation	Computes Q as reactants over products.
C	Correct	$Q = 0.20$ is below K, so more product forms.
D	Confusion set	Right comparison, reversed direction rule.

9. Answer C Concept 6 · transfer

A reaction mixture has $Q = 12$ and is observed to shift toward reactants. What must be true of K ?

Target word: reverse: from direction to K **Decides:** Reverse shift means $Q > K$.

A	Dropped element	At $Q = K$ there is no net shift.
B	Inverted operation	That would drive the reaction forward.
C	Correct	Shifting to reactants happens when Q exceeds K .
D	Absolute language	K is always positive.

10. Answer C Concept 7

A battery startup ~~that raised \$5 million~~ builds a zinc-copper galvanic cell. Standard reduction potentials are $Zn^{2+}/Zn = -0.76$ V and $Cu^{2+}/Cu = +0.34$ V. What is E°_{cell} ?

Target word: standard cell potential **Decides:** Cu is the cathode: $0.34 - (-0.76)$.

A	Dropped element	Drops the sign on the zinc potential.
B	Confusion set	Adds the two reduction potentials.
C	Correct	Cathode minus anode gives a positive, spontaneous cell.
D	Inverted operation	Swaps cathode and anode.

11. Answer A Concept 7 · transfer

A cell has $E^\circ_{\text{cell}} = +0.50$ V and transfers 2 electrons per reaction ($F = 96,485$ C/mol). What is ΔG° ?

Target word: solve for ΔG° **Decides:** $\Delta G^\circ = -nFE^\circ$.

A	Correct	$-2 \times 96,485 \times 0.50 = -96,485$ J.
B	Absolute language	Any positive E°_{cell} is spontaneous.
C	Inverted operation	Drops the minus sign; a positive E° means negative ΔG° .
D	Dropped element	Leaves out $n = 2$.

12. Answer A Concept 8

A tire filled ~~at a service station that opened in 1985~~ reads 200 kPa at 27 °C. After a highway drive the air inside warms to 57 °C with no change in volume. What is the new pressure?

Target word: new pressure **Decides:** $P_2 = 200 \times 330$ K / 300 K.

A	Correct	Kelvin ratio $330/300 = 1.1$.
B	Dropped element	Uses Celsius: $200 \times 57/27$.
C	Inverted operation	Inverts the temperature ratio.
D	Adjacent category	Adds the 30-degree change to the pressure.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owled.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	10	
Wrong stem number	A figure copied from the prompt that is not the answer.	1	
Adjacent category	The right idea in the wrong bucket.	3	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	11	
Absolute language	Always, never, guarantee — where judgment applies.	4	
Confusion set	Every option is real; the test is exact role-matching.	7	
Camouflage	Answers a stem fact that was never the question.	0	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	Gibbs free energy and spontaneity	$\Delta G = \Delta H - T\Delta S$; a process is spontaneous when $\Delta G < 0$. Convert ΔS from J to kJ before combining with ΔH .	Judging spontaneity from ΔH or ΔS alone.	ΔH , ΔS , and a temperature in one stem.
2	Hess's law	Enthalpy is a state function: add step equations; reversing a step flips the sign of its ΔH , and scaling it scales ΔH .	Adding given ΔH values without reversing the step that runs backward.	Two or more reactions combined to reach a target reaction.
3	Rate law order	Rate = $k[A]^m[B]^n$; changing $[A]$ by a factor f multiplies the rate by f^m . Orders come from experiment, not coefficients.	Assuming rate changes in simple proportion to concentration.	"Doubling ... triples/quadruples the rate."
4	Catalysts	A catalyst lowers activation energy and speeds both directions; it does not change ΔH , ΔG , or K .	Thinking a catalyst shifts equilibrium or makes more product.	"A catalyst is added ..."
5	Le Chatelier and temperature	A system at equilibrium shifts to oppose a change; for an exothermic reaction, raising temperature shifts toward reactants and lowers K .	Confusing faster rate with more product.	A sign of ΔH plus a temperature change.
6	Reaction quotient Q vs. K	If $Q < K$ the reaction proceeds forward; if $Q > K$ it proceeds in reverse; Q = products over reactants.	Inverting Q , or reversing the direction rule.	Current concentrations and a K value.
7	Cell potential and free energy	$E^\circ_{\text{cell}} = E^\circ_{\text{cathode}} - E^\circ_{\text{anode}}$; a positive E°_{cell} is spontaneous, and $\Delta G^\circ = -nFE^\circ$.	Adding the two half-cell potentials or losing the sign.	Two standard reduction potentials or an E° with electrons transferred.
8	Gas laws use kelvin	At constant volume and amount, P/T is constant with T in kelvin ($K = ^\circ\text{C} + 273$).	Plugging Celsius into a ratio.	Temperatures given in $^\circ\text{C}$ in a gas problem.

Course scope (public description): Physical Chemistry introduces the study of chemistry in terms of physical concepts. It includes thermodynamics, reaction kinetics, chemical equilibrium, electrochemistry, and matter.